

Two-way tables



- 1 Members of a gym were surveyed about what kind of training they do. Each responder only did one kind of training and the results are given in the following table:

- a Find the total number of gym members surveyed.
- b Find the number of members who do weight training.
- c Find the percentage of the gym members who do weight training.

	Cardio	Weight
Male	12	26
Female	44	18

- 2 A survey on smoking was conducted and the results are given in the table below:

- a Find the number of the people surveyed who were smokers.
- b Find the percentage of women who were nonsmokers. Round your answer to two decimal places.
- c Find the percentage of nonsmokers who were women. Round your answer to two decimal places.

	Smokers	Nonsmokers
Men	35	79
Women	56	139

- 3 The departure timing of flights out of an airport for the months of May and June are described in the given table:

- a Find the percentage of the flights in June that were delayed. Round your answer to one decimal place.
- b Find the percentage of the flights in June that were on time. Round your answer to one decimal place.
- c What fraction of the total number of flights during the two months were ones that departed on time in May?

	Departed on time	Delayed
May	144	21
June	116	40

- 4 Katrina tossed three coins. She displayed the results in the given two-way table:

- a How many times did Katrina toss heads?
- b How many more times did coin 2 land on tails than coin 1?
- c How many times did Katrina toss the coins in total?

	Heads	Tails
Coin 1	19	12
Coin 2	13	14
Coin 3	15	13

- 5 Two motels promoted packages for one night and two night stays. The following table records the results:

	One night stay	Two night stays
Motel X	463	266
Motel Y	804	817

- Find the percentage of the stays that are two night stays. Round your answer to two decimal places.
- What fraction of the two night stays was in Motel X?
- Find the percentage of the total accommodation that is in Motel X. Round your answer to two decimal places.

- 6 Michael rolled two dice. He recorded how many times each die landed on each score in the given two-way table:

Number rolled	Die 1	Die 2
1	17	13
2	9	16
3	12	7
4	10	4
5	9	6
6	7	8

- How many times did he roll a five?
- How many times did Michael roll die 2 ?
- With which die did he roll more fours?

- 7 In a biology study, the appearance of an animal when carrying a certain gene is recorded. Assume that all animals must carry Gene A or Gene B but not both. If, out of all the animals, the proportion carrying Gene A and having light fur is $\frac{4}{19}$, find the value of x .

	Gene A	Gene B
Light fur	36	37
Dark fur	28	x

- 8 500 people were asked whether they own a car or a motorcycle. The survey had the following results:

- 15% of people surveyed own a motorcycle.
- 70% of people surveyed own a car.
- 80 people surveyed do not own a car or a motorcycle.

Construct a two-way table to represent these results.

- 9 When people were asked if they found certain kinds of food addictive, the relative frequency of their responses was recorded:

	Addicted to caffeine	Not addicted to caffeine
Addicted to sugar	0.3	0.3
Not addicted to sugar	0.3	0.1

- a What proportion of responders find at least one of the two products addictive?
- b If 90 people took part in the survey, how many found at least one of the products not addictive?
- 10 Thirty six students were asked whether or not they were allergic to nuts and dairy. The two way table is provided below:

	Allergic to nuts	Not allergic to nuts
Allergic to dairy	10	6
Not allergic to dairy	6	14

- a How many students are allergic to nuts?
- b How many students are allergic to nuts or dairy, or both?
- c How many students are allergic to at most one of the two things?
- 11 In a study, some people were asked how many times they lie in a day. 20 responders said they lie at least once a day, of which 5 were children. 15 children said they never lie, and 15 adults said they never lie.

- a Complete the given table.

- b What proportion of responders said they never lie?

- c What proportion of adults said they lie at least once per day?

	0 times	1 or more times
Children		
Adults		

- 12 A survey was given to mall-goers to determine their favorite pie flavor. The results are shown in the table:

	Apple	Blueberry	Pecan
Children	124	104	70
Adults	124	42	135

To the nearest percent, what percent of the adults chose blueberry?

- 13 The following table shows the number of trains arriving either on time or late at a particular station:



Day	Arriving on time	Arriving late
Monday	23	38
Tuesday	12	25
Wednesday	22	29
Thursday	30	27
Friday	11	14
Saturday	15	11
Sunday	14	10

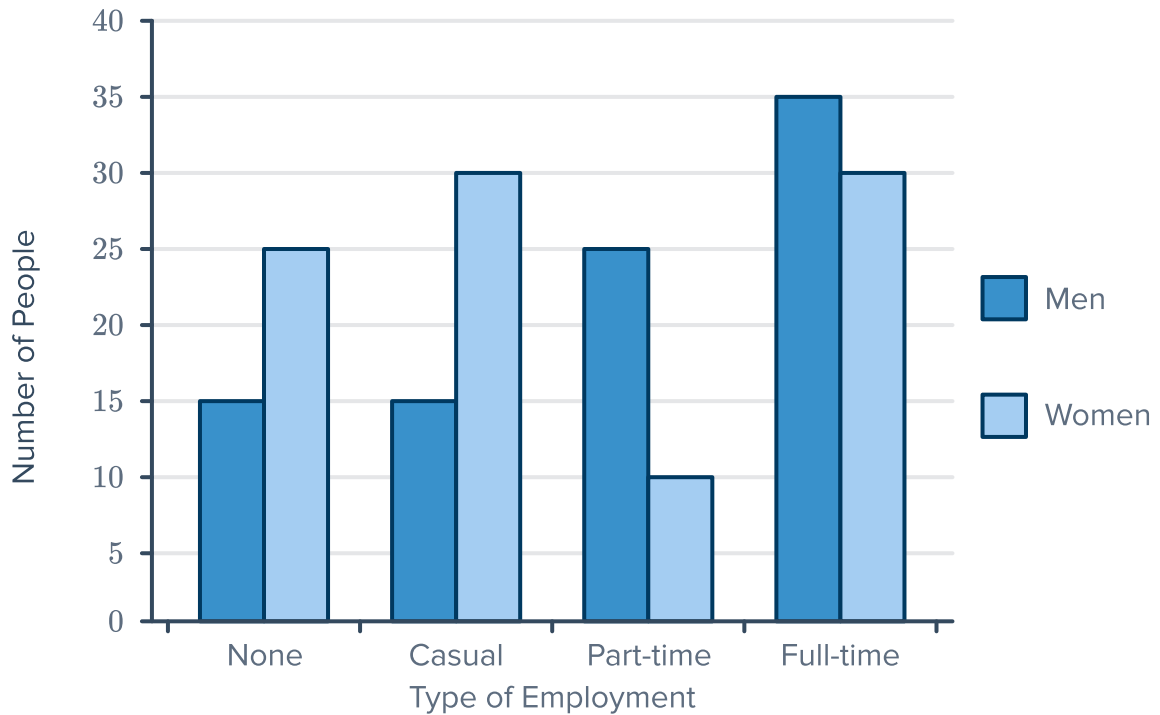
- a How many trains were late on Friday?
- b How many trains passed through the station on Wednesday?
- c How many trains were on time throughout the entire week?
- d What fraction of trains were on time over the whole week?

- 14 A total of 150 tennis players were asked whether they would support equal prize money for the women's and men's draw.

- a Find the missing value in the table.
- b How many more players are there in support of equal prize money than those against it?
- c Find the percentage of the male tennis players who support equal prize money. Round your answer to one decimal place.

	Support	Do not support
Males		35
Females	66	12

- 15 Yuri surveyed a group of people about the  of jobs they had. He recorded the data in the following graph:

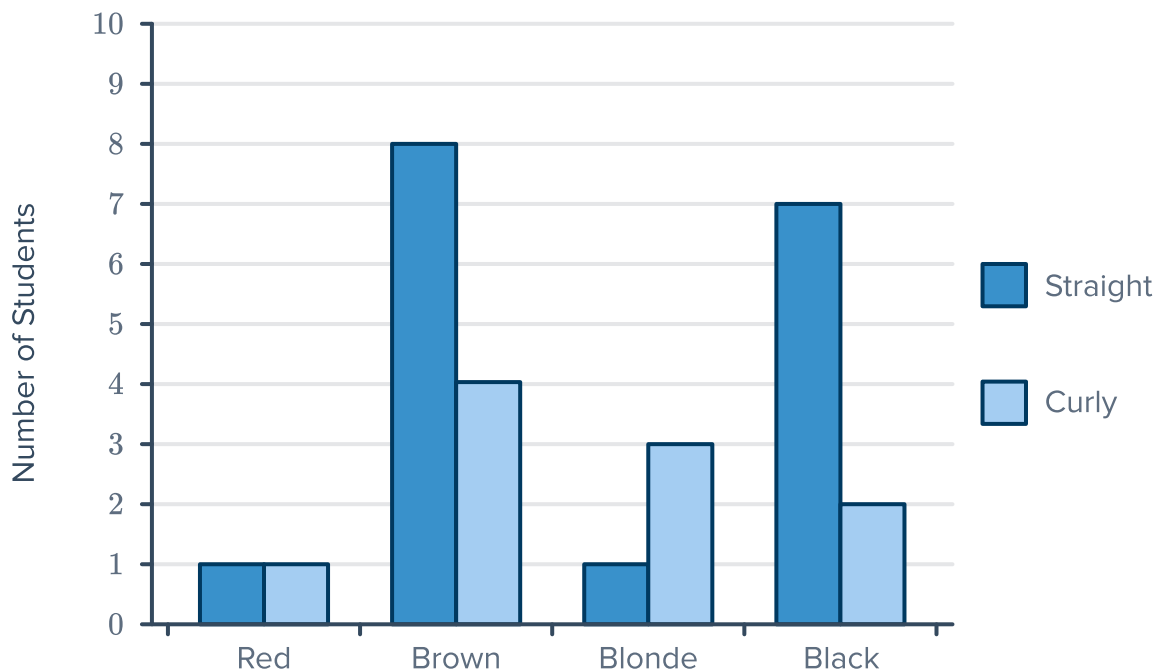


- a Complete the following two way table with the information:

	No job	Casual	Part-time	Full-time
Men				
Women				

- b How many people are unemployed?
- c What percentage of men are employed full-time? Round your answer to one decimal place.
- d What fraction of people employed casually are male?
- e If 111 people from the group are randomly selected, how many would you expect to be employed part-time?

16 Miss Merryweather recorded the hair type  color of her students in the following graph:



a Complete the following two way table with the information:

	Red	Brown	Blonde	Black
Straight				
Curly				

- b Find the number of students in the class that were surveyed.
- c What fraction of students have black hair?
- d If 15 students from the class are randomly selected, how many would you expect to have brown hair? Round your answer correct to the nearest whole number.